

Epileptic Seizure Detection from EEG Using Rank-Based Channel Selection and Neural Networks

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Abstract—Epileptic seizures (ES), characterized by sudden excessive electrical discharges in the brain, are the primary manifestation of epilepsy and often with severe consequences. Electroencephalogram (EEG), a widely adopted non-invasive technique for capturing brain activity, has become central to automated ES detection using Machine Learning. Generally, existing methods rely on data from all available EEG channels. This research focuses on a novel channel selection strategy aimed at identifying and keeping only the most informative EEG channels. The approach involves selecting EEG channels in successive stages, based on their contributions to improving accuracy, and therefore, ensures that only the most impactful channels are retained in the resulting model. Classification is conducted using Neural Networks and the model is evaluated on the CHB-MIT dataset. The model achieved a high accuracy of 98.83% while using half of the available EEG channels. Although some existing methods report narrowly higher accuracy using all channels, our approach achieves a better balance between detection performance and channel efficiency. By significantly reducing the number of required channels, the proposed method supports hardware simplification and paves the way for low-cost seizure detection systems.

Keywords— *Epileptic Seizure, Electroencephalogram, Empirical Mode Decomposition, Channel Selection, Neural Network*

I. INTRODUCTION

Epileptic seizures (ES) are sudden disruptions in brain activity caused by abnormal electrical discharges in the brain, affecting behavior, consciousness, or motor control [1]. EEG is the most widely used non-invasive tool for detecting these irregularities, offering high temporal resolution of neural signals. The distinct temporal and spectral patterns associated with seizures make EEG a valuable resource for automated seizure detection. However, the chaotic, non-stationary nature of EEG signals presents challenges in distinguishing seizure events from normal brain activity.

Epilepsy has been an active research area within computational intelligence and biomedical signal processing for several decades, with significant efforts directed toward the development of automated ES detection systems using EEG data ([2], [3]). EEG is considered the most reliable and widely used signal for seizure detection due to its high temporal resolution, non-invasive nature, and affordability [4]. With the advancement of Machine Learning (ML) and

Deep Learning (DL), recent studies have explored EEG-based diagnosis not only for epilepsy but also for neurological conditions such as autism spectrum disorders and emotion recognition ([5], [6]). Seizure events in EEG are typically characterized by sudden bursts of abnormal rhythmic activity, such as spikes, sharp waves, or spike-and-wave complexes. However, the detection process is complicated by the nonlinear, non-stationary nature of EEG signals, the presence of noise during recording, and the high similarity between seizure and non-seizure patterns in terms of frequency and amplitude. These challenges have motivated researchers to design robust preprocessing, feature extraction, and classification techniques capable of generalizing across subjects and varying seizure types.

Recent EEG-based ES detection studies have investigated a range of strategies for signal processing, feature extraction, and classification using ML and DL techniques. Several works have utilized raw EEG signals by directly applying DL models with built-in feature extraction capabilities [7]. In parallel, other studies have focused on extracting handcrafted or statistical features from EEG to enhance discriminative power ([8], [9]). A growing body of research emphasizes signal transformation and feature engineering to improve recognition performance, including the use of decomposition methods and filtering techniques for better signal representation ([8], [10]).

The study focuses on optimizing EEG channel selection for ES detection in a patient-independent framework. Building on previously established feature extraction methods [11], this work prioritizes identifying the most informative subset of EEG channels to reduce computational complexity and facilitate practical deployment. Each of the EEG channels is independently evaluated, and channels are ranked based on their performance. For the ES system with selected channels, rank-based channel selection is investigated, where channels are checked following the order of their rank to choose for the final model. The proposed approach ensures that only the most relevant channels are selected, resulting in a compact yet effective ES detection system. The method significantly reduces the number of required EEG channels while maintaining high classification accuracy, making it well-suited for efficient, scalable, and real-world seizure detection systems.

In the rest of this paper, the proposed model and its related research are introduced in detail. Section II provides a brief overview of existing ES detection studies as a reference of the current study. Section III introduces the proposed methodology in detail, covering EEG data preparation, signal decomposition and feature extraction, the rank-based channel selection process, and classification with NN. Section IV presents the experimental results and analysis, including experimental setup and a comparative evaluation with existing related state-of-the-art methods. Finally, Section V concludes the paper highlighting the key findings of the study and outlining potential future research directions of this study.

II. RELATED WORKS

Various ML techniques have been applied to multichannel signal classification tasks. A large portion of existing studies utilize the full set of available EEG channels without channel-specific selection or reduction. Pattnaik et al. [8] employed the Tunable Q-wavelet Transform (TQWT) for signal decomposition, followed by the extraction of nonlinear, temporal, and statistical features and classified using Support Vector Machine (SVM) and Random Forest (RF) algorithms. Nandini et al. [12] utilized two different time-domain feature extraction techniques, and a range of classifiers, including SVM, K-Nearest Neighbors (KNN), Logistic Regression (LR), RF, Decision Tree (DT), and Naive Bayes (NB) for evaluating the task of seizure detection. Salafia et al. [13] explored the use of Mutual Information (MI) to assess inter-channel correlations and applied a 2D CNN for seizure detection. This approach bypassed explicit feature extraction by allowing the CNN to learn directly from the EEG data. He et al. [3] applied Graph Attention Networks (GAT) to capture spatial relationships within the EEG data, while Bi-directional Long Short-Term Memory (BiLSTM) networks were used to detect epileptic seizures directly from raw EEG signals.

Some other works also investigated the full set of EEG channels. Hassan et al. [4] proposed an epileptic seizure detection method combining Empirical Mode Decomposition (EMD), MI-based feature selection, and a multi-layer perceptron NN (MLPNN). The Mutual Information-based Best Individual Feature (MIBIF) algorithm was then applied to select features before classification using MLPNN. Liu et al. [14] proposed a novel feature-based seizure detection algorithm leveraging clinically and statistically significant features extracted from EEG signals. Al-Adhaileh et al. [15] introduced a multi-model algorithm for classifying epileptic seizure conditions using EEG signals. The study processed data from 102 patients, extracting frequency-band (delta to gamma) and entropy features before applying SMOTE for class balancing. They compared several ML and DL models on different datasets. Das et al. [11] proposed a seizure detection method using 1D and 2D features from EMD-decomposed EEG signals, extracting fluctuation index, variance, and ellipse area from IMFs. The 2D-CNN with image-like features outperformed the 1D-CNN. Additionally, in [16], a seizure detection method was developed using an integrated feature set from EMD-decomposed EEG signals.

In contrast to using all the channels, a few recent studies focus on reducing the feature space or selecting the most informative EEG channels to improve interpretability and reduce computational complexity, while still maintaining high classification performance. Recent research has shifted toward channel selection strategies that aim to retain only the most informative signals. Ferrara et al. [17] proposed a seizure detection by Principal Component Analysis (PCA) based

channel ranking to identify two patient-specific channels from standard multi-electrode recordings and trains a CNN on these selected channels. Dokare and Gupta [18] introduced a seizure detection method that combines MI and RF for selecting relevant EEG channels, followed by DWT to extract ten band-specific features. These are classified using an SVM with CHB-MIT dataset. Moctezuma and Molinas [19] investigated EEG channel selection for seizure detection using a multi-objective optimization approach based on NSGA-II and NSGA-III. Their method combines EMD or DWT for signal decomposition with energy and fractal dimension feature extraction, aiming to maximize accuracy while minimizing the number of selected channels.

Most existing studies on EEG-based ES detection adopt patient-specific channel selection strategies, where a small subset of channels, typically two or four or more, is chosen individually for each patient. While this approach can yield high accuracy within a specific case, the selected channels often vary across individuals and lack consistency, making them unsuitable for generalized models. This patient-dependent nature limits the scalability and real-world applicability of such methods, especially in scenarios requiring uniform channel configurations, such as wearable or clinical monitoring systems. Motivated by this gap, the present study aims to develop a validation-driven, performance-based channel selection method that identifies a common set of informative channels capable of generalizing well across patients, without the need for patient-specific tuning.

III. EPILEPTIC SEIZURE DETECTION FROM EEG USING RANK-BASED CHANNEL SELECTION

This study proposes a systematic approach for a suitable ES detection system from EEG signals while aiming to consider reducing the number of channels (i.e., electrodes). The proposed framework is depicted in Fig. 1. The whole methodology comprises preprocessing of EEG signals, signal decomposition, feature extraction, proposed rank-based channel selection, and measuring performance (i.e., classification accuracy) with NN and CNN. The following sub-sections briefly describe major components of the proposed ES detection system.

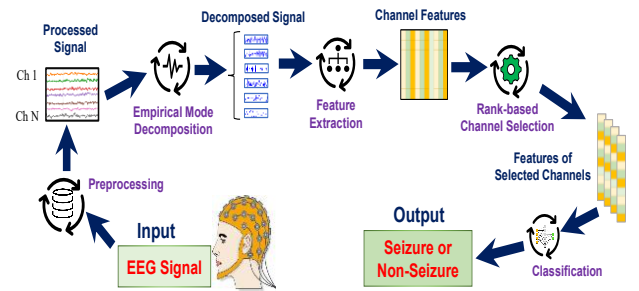


Fig. 1. Framework of Proposed Epileptic Seizure Detection from EEG Signal using Rank-based Channel Selection.

A. EEG Data Preprocessing

EEG data used in this study were obtained from the publicly available CHB-MIT Scalp EEG Database [20], which was used in several recent ES detection studies ([17], [18]). The dataset contains long-term EEG recordings from 24 pediatric epilepsy patients. A total of 22 EEG channels were considered in this study from 24 cases. Each EEG signal was segmented using a fixed-length sliding window of 10 seconds with 70% overlap, which effectively results in a 3-second step

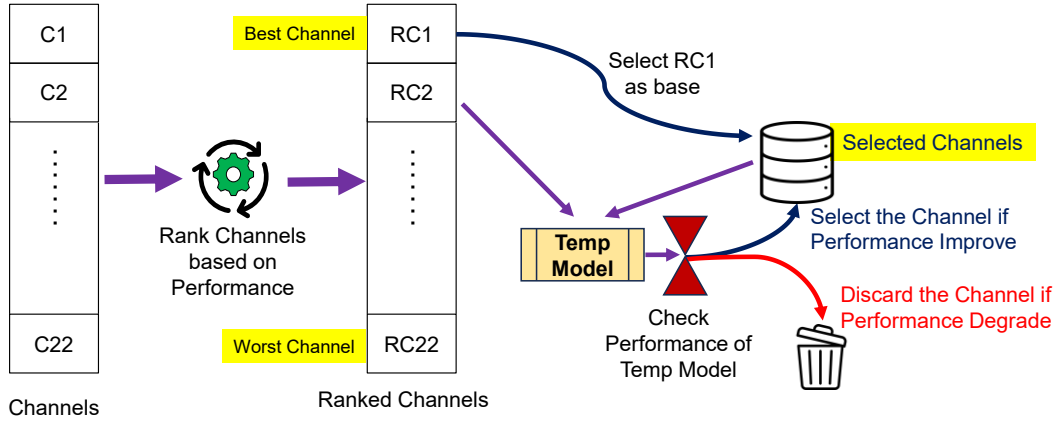


Fig. 2. Rank-based Channel Selection (RCS).

size between segments. Each segmented window was then labeled as seizure or non-seizure based on the annotations provided in the dataset, resulting in a total of 7414 labeled data samples used in the experiments.

B. Empirical Mode Decomposition (EMD)

To handle the nonlinear and non-stationary nature of EEG signals, each windowed segment was decomposed into its constituent components using EMD. EMD adaptively breaks down each EEG signal into a finite number of IMFs, which represent oscillatory modes at various time scales inherent to the data [21]. For this study, the first six IMFs were retained from each segment, as they contain most of the relevant frequency components for seizure detection ([11], [16]). The decomposition was performed independently for each of the 22 EEG channels.

C. Channel-wise Feature Extraction

Following decomposition, a set of handcrafted features was extracted from each of the six IMFs per EEG channel. For each IMF, three types of features were computed: (i) Variance (V), representing signal energy; (ii) Fluctuation Index (F), capturing signal instability and fluctuation magnitude; and (iii) Ellipse Area (EA) of the second-order difference plot, quantifying the geometric dispersion of the signal in a nonlinear phase space. These features are well-established in the literature for capturing temporal and nonlinear dynamics of EEG signals [22]. In total, each EEG channel produced 18 features (6 IMFs $\times$ 3 features). Thus, the dimensionality of the feature vector for a single channel was 18, for two channels 36, and so on. One may go through the existing studies [11], [16] for a detailed description of the feature extraction from the decomposed signal.

D. Rank-based Channel Selection (RCS)

Selection of the most informative EEG channels and ES detection with appropriately selected channels is the main objective of the study. Rank-based channel selection (RCS) is the innovative but simple idea introduced in this study to choose channels for well-performed ES detection model development. RCS follows straightforward steps: evaluate the performance of each individual channel, rank channels based on performance, and finally select channels according to rank. Fig. 2 depicts the major steps of RCS. The performance of individual channels is measured by considering individual models for each channel. Ranking individual channels (i.e., RC1 to RC22) by performance (i.e., accuracy) is also straightforward. The selection of channels for the final model from the ranked channels is the crucial step of RCS. Initially, RC1 is selected as the base performance channel for the model. To consider other channels (RC2 to RC22), channels are checked one by one according to rank. A channel is added

to the model with the selected channels and a temporary model. If the temporary model outperforms the model with already selected channels, the channel is considered for the model; otherwise, it is discarded. The selection process progresses until RC22, and the outcome RCS is the model of the selected channels. Remarkably, a channel consideration for the model is the inclusion of 18 features generated from the channel, along with features from already chosen channels. The temp model is trained on the training set, but its performance is measured using a different set (i.e., the validation set in this study).

E. Classification with NN

ES detection from the CHB-MIT dataset is a binary classification task (i.e., Seizure or Non-Seizure), and the proposed classifier system is developed with NN. Two different NN models with single hidden layer having 10 and 50 Hidden Neurons (HN) are considered. Similar models are also considered in the existing studies, where detailed descriptions are available [11], [16]. Both the models were developed using the same data partitions of training and validation sets; performance was evaluated using a test set that was reserved from the available data. Model comparisons were conducted both in terms of final test accuracy (i.e., generalization) and the number of channels selected for a model.

IV. EXPERIMENTAL STUDIES

The primary goal was to find a minimal subset of channels that preserves classification accuracy that is important for reducing hardware/electrode requirements. We performed the rank-based channel selection, i.e., RCS: each channel was first evaluated individually (using validation accuracy), and then channels were sorted by their respective performance. The channel selection was performed with two classification models: an NN with one hidden layer having 10 neurons (NN-10), a similar NN with 50 neurons (NN-50). For each model, channels were incrementally added based on a greedy selection strategy that prioritized validation accuracy improvement at each step.

To evaluate the effectiveness of the proposed RCS approach, the CHB-MIT scalp EEG database was used. From each EEG channel's segment, we computed 18 features, so that adding a channel increases the feature dimension by 18 (for instance, one channel = 18 features, two channels = 36 features, etc.). There were 7414 samples. The experiments, including feature extraction and classification, are conducted using Python. To experiment, 20% of the 7414 samples were reserved as a test set, which was not used in any model development process and was only used to measure the generalization of the developed model. Among the 80% of the

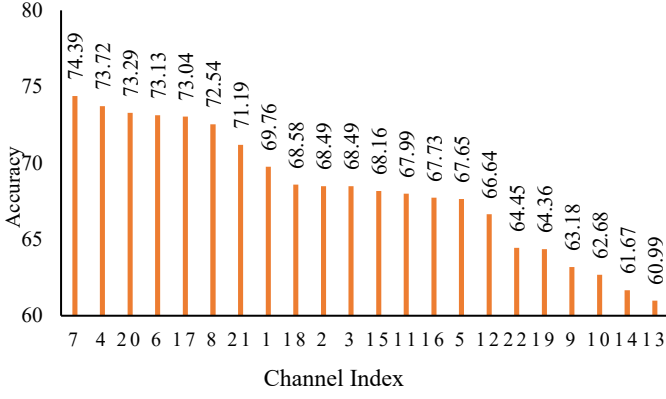


Fig. 3. Accuracy achieved by NN-10 model trained on individual EEG channels.

samples, 60% were used to train a model, and 20% were used as a validation set to measure the model's intermediate performance.

A. Performance with Individual Channel

The histogram in Fig. 3 illustrates the validation accuracy achieved by every individual EEG channel (1 to 22) independently using NN-10 for ES detection. Channels are presented according to accuracy rank in accuracy percentage which ranges from 60% to 75%. It is found that channel 7 is the best, with an accuracy of 74.39, and channel 13 is the worst, with an accuracy of 60.99. Notably, channels such as 4, 6, 7, 8, 17, 20, and 21 exhibit higher validation accuracies, indicating their strong potential in capturing discriminative features related to seizure activity. In contrast, channels like 10, 11, 18, and 22 show comparatively lower performance, suggesting limited contribution to classification when used alone. The results highlight that EEG channels vary in their effectiveness for seizure detection, reinforcing the importance of channel selection strategies, and a model with only the most informative channels might enhance overall model performance.

B. Result Analysis

Starting from the best individual channel (i.e., channel 7 with 74.39% validation accuracy), the successive addition of channels is investigated for NN-10 and NN-50 models. Table I shows the NN-10 model's validation accuracy considering the ranked channels shown in Fig. 4, i.e., 7, 4, 20, 6, 17, 8, 21, 1, 18, 2, 3, 15, 11, 16, 5, 12, 22, 19, 9, 10, 14, and 13. In the model, channel 7 is selected as the baseline because it is the best individual channel. The addition of channels 4, 20, and 6 improves model performance (as shown in the Upward Arrow), and the validation accuracy with channels 7, 4, 20, and 6 is 85.26%. In contrast, adding channels 17 and 8 degraded the model's performance (as shown in the downward arrow); therefore, we discarded these channels and considered the next channel, 21, for the model. At a glance, the model's validation accuracy is 96.97% with 12 selected channels (i.e., 7, 4, 20, 6, 21, 1, 3, 15, 16, 5, 19, and 14) while discarded 10 channels (i.e., 17, 8, 18, 2, 11, 12, 22, 9, 10, and 13).

Table II shows the NN-50 model's performance in a similar manner to the channel consideration shown in Table I for NN-10. It began with channel 7 (having validation accuracy of 78.22%), and improved performance more significantly with additional channels. Ultimately, the highest validation accuracy of 98.56% was achieved using 11 selected channels (i.e., 7, 4, 20, 6, 17, 8, 21, 1, 3, 15, and 9) while discarded other 11 channels (i.e., 18, 2, 11, 16, 5, 12, 22, and

19). The NN-50 model outperforms NN-10, even with a smaller number of channels. The improved performance with fewer channels underscores how increased model capacity enables better feature utilization, making NN-50 significantly more channel-efficient than NN-10. Again, such observation suggests a deeper network architecture improved accuracy but required careful channel selection to minimize redundancy.

TABLE I. PERFORMANCE OF NN-10 MODEL WITH SELECTED CHANNELS.

Check Channel	Validation Accuracy	Comment	Channels in the Model
7	74.39	Selected as Baseline	7
4	78.85 ↑	Selected	7, 4
20	83.49 ↑	Selected	7, 4, 20
6	85.26 ↑	Selected	7, 4, 20, 6
17	85.03 ↓	Discarded	7, 4, 20, 6
8	80.23 ↓	Discarded	7, 4, 20, 6
21	86.60 ↑	Selected	7, 4, 20, 6, 21
1	92.08 ↑	Selected	7, 4, 20, 6, 21, 1
18	91.98 ↓	Discarded	7, 4, 20, 6, 21, 1
2	91.90 ↓	Discarded	7, 4, 20, 6, 21, 1
3	92.33 ↑	Selected	7, 4, 20, 6, 21, 1, 3
15	92.67 ↑	Selected	7, 4, 20, 6, 21, 1, 3, 15
11	92.50 ↓	Discarded	7, 4, 20, 6, 21, 1, 3, 15
16	92.67 ↑	Selected	7, 4, 20, 6, 21, 1, 3, 15, 16
5	93.34 ↑	Selected	7, 4, 20, 6, 21, 1, 3, 15, 16, 5
12	92.60 ↓	Discarded	7, 4, 20, 6, 21, 1, 3, 15, 16, 5
22	93.08 ↓	Discarded	7, 4, 20, 6, 21, 1, 3, 15, 16, 5
19	95.20 ↑	Accepted	7, 4, 20, 6, 21, 1, 3, 15, 16, 5, 19
9	95.11 ↓	Discarded	7, 4, 20, 6, 21, 1, 3, 15, 16, 5, 19
10	94.30 ↓	Discarded	7, 4, 20, 6, 21, 1, 3, 15, 16, 5, 19
14	96.97 ↑	Selected	7, 4, 20, 6, 21, 1, 3, 15, 16, 5, 19, 14
13	95.90 ↓	Discarded	7, 4, 20, 6, 21, 1, 3, 15, 16, 5, 19, 14

TABLE II. PERFORMANCE OF NN-50 MODEL WITH SELECTED CHANNELS.

Check Channel	Validation Accuracy	Comment	Channels in the Model
7	78.22	Selected as Baseline	7
4	88.67 ↑	Selected	7, 4
20	93.86 ↑	Selected	7, 4, 20
6	96.42 ↑	Selected	7, 4, 20, 6
17	96.83 ↑	Selected	7, 4, 20, 6, 17
8	97.10 ↑	Selected	7, 4, 20, 6, 17, 8
21	97.52 ↑	Selected	7, 4, 20, 6, 17, 8, 21
1	98.04 ↑	Selected	7, 4, 20, 6, 17, 8, 21, 1
18	97.64 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1
2	97.64 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1
3	98.44 ↑	Selected	7, 4, 20, 6, 17, 8, 21, 1, 3
15	98.51 ↑	Selected	7, 4, 20, 6, 17, 8, 21, 1, 3, 15
11	97.47 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15
16	97.47 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15
5	97.30 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15
12	97.13 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15
22	97.89 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15
19	97.81 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15
9	98.56 ↑	Selected	7, 4, 20, 6, 17, 8, 21, 1, 3, 15, 9
10	96.29 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15, 9
14	96.96 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15, 9
13	97.81 ↓	Discarded	7, 4, 20, 6, 17, 8, 21, 1, 3, 15, 9

Figure 4 demonstrates the accuracy on the test set along with the validation set for both NN models, for a better understanding of the selection of channels based on the

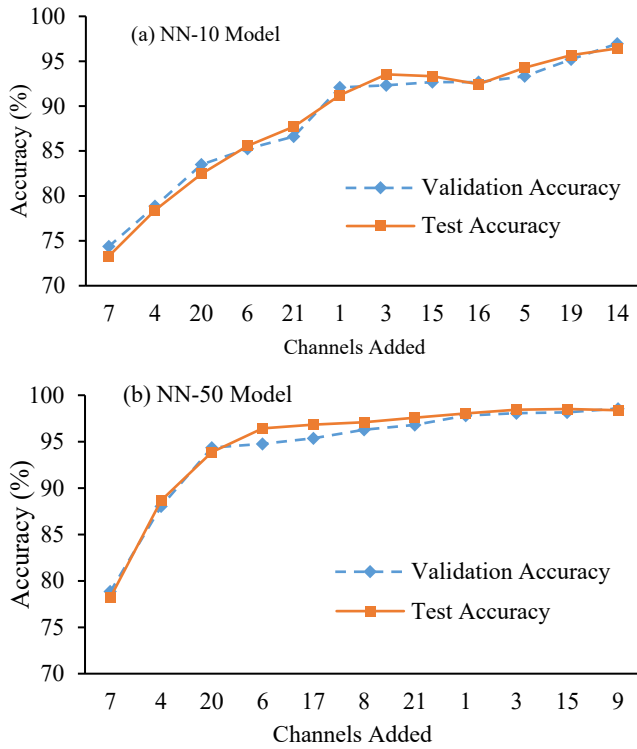


Fig. 4. Accuracy of validation and test sets on NN-10 and N-50 models while added EEG channel selectively.

validation set. Performance on the sets is shown to improve with selection of channels. Between the NN-10 and NN-50 models, the NN-50 model shows a better improvement in Fig. 4(b) with a smaller number of channels compared to the NN-10 model. With three channels, the test set accuracies for NN-10 and NN-50 models are 80.47% and 93.86%, respectively. Test set performance with 10 channels for NN-50 is 98.52% and it is 95.68% for NN-10. The NN-50 model, which included a larger hidden layer, demonstrated markedly better efficiency.

Table III shows the summary of the outcomes for both the models. The experimental results indicate that validation accuracy closely aligns with test accuracy trends, validating the effectiveness of the RCS method. Between the models, NN-50 demonstrated the highest overall performance and channel efficiency. Adding channels selectively, the optimal validation set accuracy of 98.57% was obtained by NN-50 using 11 selected channels (i.e., 7, 4, 20, 6, 17, 8, 21, 1, 18, 2, 3), and the test accuracy peaked at 98.38%.

TABLE III. OUTCOMES OF NN-10 AND NN-50 MODELS WITH SELECTED CHANNELS.

Model	Selected Channels with RCS	Number of Channels	Max Validation Accuracy Raised (%)	Test Accuracy (%)
NN-10	7, 4, 20, 6, 21, 1, 3, 15, 16, 5, 19, 14	12	96.97	96.43
NN-50	7, 4, 20, 6, 17, 8, 21, 1, 18, 2, 3	11	98.57	98.38

Another crucial observation from Table III is that across all models, there was the consistent selection and significance of particular EEG channels. Channels 4, 6, 7, 20 and 21 notably appeared in the selection process for both the models, indicating their substantial influence on seizure detection performance. The recurrent presence of these specific channels across diverse architectures underscores their potential as primary EEG leads in practical implementations,

enhancing both effectiveness and efficiency. Thus, careful selection centered on these consistently informative EEG channels could lead to substantial hardware simplification, cost reduction, and improved portability.

C. Performance Comparison with Existing Studies

Table IV compares the performance (i.e., test set accuracy as a generalization ability) of the proposed method with that of prominent existing EEG-based epileptic seizure (ES) detection methods with the CHB-MIT dataset. Data segmentation strategy, decomposition and feature engineering techniques, channel selection methods and classification models are also presented in the table for better understanding. The results of the existing studies are reported in the corresponding articles. Several prominent existing studies did not consider channel selection; therefore, their performance is based on all 22 channels. In this category, Hassan et al. [4] and Kong et al. [23] combined decomposition techniques like EMD and DWT with handcrafted features and machine learning classifiers, and achieved accuracy of 99.57% and 99.32%, respectively. Das et. al [16] employed EMD-based features and an NN model without channel selection, reporting top-tier accuracy close to 99.8%. The test set accuracy of the proposed method with 11 selected channels is found to be competitive with the existing prominent methods with all 22 channels.

TABLE IV. COMPARISON OF THE PROPOSED METHOD WITH NOTABLE EXISTING APPROACHES ON ES DETECTION USING CHB-MIT DATASET.

Author [Ref., Year]	Train-Test Split	Decomposition + Feature Extraction (FE) + Feature Selection (FS)	Channel Selection (CS) Method: Number of Selected Channels	Classification using ML/DL	Achieved Test Set Accuracy (%)
Hassan <i>et al.</i> [4], 2022	K-fold cross-validation (CV) K = 2, 5, 10	EMD + Ellipse area, variance, Fluctuation Index + MI	[No CS]: 22	MLPNN	99.57
Mohamm adpoory <i>et al.</i> [24], 2025	Leave-One-Patient-Out (24-fold CV)	DWT + Weighted Visibility Graph	[No CS]: 22	RF, SVM, KNN, NB, LDA (Best: RF with RFE-P features)	94.02
Kong <i>et al.</i> [23], 2024	10-fold CV	DWT + Welch + PMPSO	[No CS]: 22	SVM, ANN, RF and XGBoost	99.32
S. Das <i>et al.</i> [16], 2024	80-20	EMD + Fluctuation Index, Variance, Ellipse area + [No FS]	[No CS]: 22	NN	99.80
Dokera and Gupta [18], 2025	70/30 split, then on 70% 5-fold CV	DWT + [No FE] + [No FS]	MI and RF: Channel number varies with patient	SVM	97.70
Moctezuma <i>et al.</i> [19], 2020	10-fold CV	EMD/DWT + [No FS]	Backward-elimination algorithm and NSGA-III: 1 to 3	SVM, KNN, RF	1 Ch.: 97 2 Ch.: 98 3 Ch.: 99
Proposed Method	80-20	EMD + Fluctuation Index, Variance	Rank-based Selection: 11	NN	98.38

It is clearly observed from Table IV that the proposed method outperformed existing channel selection-based methods. It is a critical matter to select appropriate channels without sacrificing model accuracy. In general, there is a trade-off between model complexity, number of EEG channels, and model performance. Retaining the model's generalization (measured on the test set) with the selected channel is a crucial issue, as channel selection/discard is typically performed using the training/validation sets. Moctezuma *et al.* [19] employed evolutionary optimization-based channel selection strategies, which appear complex, significantly reducing the number of channels (down to 1-3) while maintaining reasonable accuracy (97% to 99%) on 10% test data (as 10-fold cross-validation). The recent method, Dokera and Gupta [18], considered MI and RF for channel

selection and achieved 97.70%. In contrast, the proposed method is a rank-based channel selection method, which is straightforward to identify a compact and effective subset of channels. The proposed method achieved a competitive accuracy of 98.38% with 11 channels using a NN model, demonstrating a strong trade-off between performance and channel efficiency. This comparison highlights the importance of balanced channel selection mechanisms for building clinically viable, computationally efficient seizure detection systems.

V. CONCLUSIONS

In this study, a comprehensive pipeline was proposed for epileptic seizure (ES) detection with an emphasis on minimizing the number of EEG channels while maintaining high classification accuracy. CHB-MIT recordings were segmented into overlapping 10s windows, decomposed with EMD to obtain the first six IMFs, from which three features per IMF were extracted, yielding an 18-dimensional feature vector per channel. A novel rank-based channel selection procedure, initially guided by the single-channel performance of a simple neural network and subsequently validated across more complex feedforward architectures, was employed to identify an effective subset of the channels. The rank-based channel selection subsequently and transparently identifies the smallest subset of channels (i.e., electrodes) that preserves high classification accuracy. This subset achieved near-optimal accuracy when evaluated on held-out test data. The experimental findings achieved test accuracies in the range of approximately 98.38% using half of the total channels. These results provide strong evidence that appropriate channel selection strategy can effectively maintain high classification accuracy with fewer channels, thereby supporting the development of lightweight and portable EEG monitoring systems for ES detection.

Future research may extend this work in several meaningful directions. In this study, EMD is used to decompose EEG signal and no feature selection was considered. Other decomposition methods (e.g., Discrete Wavelet Transform) with channel selection can be explored for better ES detection. Furthermore, the rank-based channel selection strategy is simple, straightforward and interpretable but not fully explore all possible channel combinations; therefore, the integration of advanced optimization techniques may yield improved channel selection outcomes and which could be considered a challenging extension for future work.

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